KakeyaLattice: Nested-Lattice KV-Cache Compression with Kakeya-Style Discrete Codebooks

Allen Li
Individual researcher
Email: AllenL329@gmail.com

April 24, 2026

Abstract

We introduce KakeyaLattice, a KV-cache compression codec for transformer-based large language models that constructs a *discrete Kakeya set* over the direction sphere via a Zamir–Feder nested-lattice quantiser [24]. The paper covers two concrete lattice instantiations of a single codec family: a D_4 *nested lattice* variant and an E_8 *nested lattice* variant. Both share the same nine-step pipeline and differ only in the choice of shaping lattice. In the classical continuous setting a Kakeya set contains a unit segment in every direction; we replace "continuous unit segments" by "lattice Voronoi cells scaled by a per-vector adaptive range", so the codebook is an *explicit discrete covering of* S^{D-1} rather than a scalar-quantisation grid.

The central innovation of the D_4 nested lattice design is adaptation to the non-Gaussian distribution of real LLM KV activations. Measured on Qwen3-4B post-QK-norm K, the activations are sub-Gaussian in the body (excess kurtosis < 3), heavy-tailed per-coordinate after Hadamard whitening (RMS Wasserstein-$2/\sigma = 0.65$), and anisotropic (variance ratio up to $4.71\times$ across coordinates) — a distribution for which the implicit i.i.d. Gaussian assumption of every scalar quantiser is a poor fit. The D_4 nested lattice variant addresses each of these structural mismatches explicitly: unit-norm factorisation removes the per-vector magnitude spread; a Sylvester–Hadamard rotation whitens coordinate anisotropy; a per-vector adaptive $q_{\max}$ absorbs the heavy per-vector tail; and the D_4 Voronoi cells replace the cube cells of $\mathbb{Z}^4$, trading $G(\mathbb{Z}^4) = 1/12$ for the unique second-moment optimum $G(D_4) \approx 0.0766$ in $\mathbb{R}^4$ [26]. The E_8 nested lattice variant takes the same pipeline and upgrades the lattice from D_4 to E_8, whose Voronoi region is closer still to a sphere: $G(E_8) \approx 0.0717$, giving a further $+0.29$ dB of shaping gain over D_4 at matched rate.

The D_4 nested lattice effect is measured directly: on 12 head-to-head pairs (four open-source models $\times$ three matched bit points) the D_4 nested lattice variant wins universally on K-MSE (12/12, 10–36% better) and 9/12 on $|\Delta\text{ppl}|$ (4/4 at the aggressive $\sim 3.6\times$ compression point with 1.8–5.3$\times$ improvement). The E_8 nested lattice upgrade adds a further $+1.3$ to $+2.0$ dB per-layer K-MSE gain across Qwen3-4B, DeepSeek-R1-Distill-Qwen-1.5B, Gemma-4-E4B, and GLM-4-9B-Chat — 4–6$\times$ the $+0.29$ dB theoretical minimum, because E_8's two-coset structure handles coarse-quantisation outliers better than D_4's single-flip parity correction. In real vLLM in-forward no-boundary $|\Delta\text{ppl}|$ at $n = 32$ passages with 95% CI, the E_8 nested lattice variant reduces $|\Delta\text{ppl}|$ by 28–53% across three deployable models at both $Q = 4$ (aggressive) and $Q = 10$ (balanced) operating points. The cost is a fixed $+32$ bits per vector (-5–7% compression ratio vs the D_4 nested lattice variant at iso-Q) and 1.56–1.80$\times$ encode latency (two D_8 candidate cosets + distance compare instead of one parity flip). Long-context retrieval (Needle-in-a-Haystack at 16k context) is preserved on Qwen3-4B and Gemma-4-E4B; GLM-4-9B shows a mild regression (89% vs 100%) at $Q = 10$ on the E_8 nested lattice variant, attributable to a RoPE/Voronoi interaction discussed in the caveats.

All numbers are live-vLLM [48], strict-GPU on H200, no mock / no simplification / no fallback / no overfit. Bit-level regression protection is enforced by a pinned `sha256` frozen-parity test: 8 hashes (D_4 and E_8 nested lattice variants at four Q values each) assert byte-identical codec output against the release-tag snapshots. Code, per-passage JSON, four

per-architecture attention hooks (Qwen3Attention, Qwen2Attention, Gemma4Attention, GLMAttention), and the multi-model / NIAH / latency harnesses are released under Apache-2.0.

1 Introduction

The key–value (KV) cache is the dominant memory constraint in production LLM inference [42, 43, 47]. At context length T, a model with L layers, H_{kv} key/value heads, and head dimension d stores $4 \cdot L \cdot H_{kv} \cdot d \cdot T$ bytes in bf16. For Qwen3-4B at $T = 128{,}000$ this is 18 GiB per user; for the April-2026 flagship open-source models (DeepSeek-V3.1, Kimi-K2, Qwen3-235B, GLM-4.6) per-user footprints at 128k are several hundred GiB each. Compressing this object without re-training is therefore one of the highest-leverage algorithmic targets in current serving infrastructure.

1.1 Why lattice shaping matters

Scalar quantisation after an orthogonal rotation is the core pattern of essentially every KV codec: a rotation whitens coordinate-wise correlations, and per-coordinate Lloyd–Max or uniform quantisation applies to the rotated residual. Shannon's rate–distortion theory gives a tight lower bound on such codecs: the per-coordinate mean-squared reconstruction error of an N-dimensional scalar quantiser at rate b bits per coordinate is bounded below by $G(\mathbb{Z}^N) \cdot 2^{-2b}$, where $G(\mathbb{Z}^N) = 1/12$ is the normalised second moment of the cube [24, 26]. For a generic N-dimensional lattice Λ, replacing the cube by the Voronoi cell of Λ gives

$$\mathrm{MSE}_\Lambda \;=\; G(\Lambda) \cdot 2^{-2b} \quad \text{with} \quad G(\Lambda) \le G(\mathbb{Z}^N), \tag{1}$$

with equality iff $\Lambda = \mathbb{Z}^N$. The ratio $G(\mathbb{Z}^N)/G(\Lambda)$ (in dB) is the *shaping gain* of Λ over $\mathbb{Z}^N$. Known optimal values in low and moderate dimensions:

Dimension	Lattice	$G(\Lambda)$	Shaping gain over $\mathbb{Z}^N$
1	$\mathbb{Z}$	$1/12 = 0.0833$	0.00 dB
2	A_2	0.0802	0.17 dB
4	D_4	0.0766	**0.37** dB
8	E_8	0.0717	0.65 dB
24	Λ_{24}	0.0658	1.03 dB

No currently deployed KV codec exploits this shaping gain: every production codec of which we are aware quantises after rotation *per coordinate* (the $\mathbb{Z}^N$ case). This leaves between 0.37 dB and 1.53 dB of unconditional MSE reduction on the table at matched bit budget — a strictly free improvement, constrained only by the encode-decode cost of the lattice.

1.2 The codec as a discrete Kakeya set

The classical Kakeya problem [1, 2, 3] asks for the minimum Hausdorff-dimensional measure of a set in $\mathbb{R}^D$ containing a unit segment in every direction. Its role as a touchstone of harmonic analysis [4, 5, 6, 7, 8, 10] is well documented; the finite-field analogue is resolved by Dvir's polynomial method [11], and the three-dimensional Euclidean case was resolved in 2025 by Wang and Zahl [21].

KakeyaLattice's codebook is a *discrete Kakeya set* constructed explicitly on the direction sphere S^{D-1}: every codeword is the image under an orthogonal rotation and unit-norm rescaling of a D_4 lattice point, and the tensor product $D_4^{\otimes D/4}$ of the lattice's Voronoi cells tiles $\mathbb{R}^D$ so that every direction in $\Theta = \{x_i / \|x_i\|\}$ lies within one cell. The classical Kakeya set's "unit

segment in every direction" is replaced, at scale $q_{\max}/q_{\text{range}}$ set adaptively per vector, by "D_4 Voronoi cell scaled to the vector's infinity norm"; the unconditional sphere-packing optimality of D_4 in $\mathbb{R}^4$ [26] gives this cover its optimal second-moment per bit. Proposition 2.2 (§2) makes this correspondence quantitative.

The construction is linked to the chain multilinear Kakeya [14] → Brascamp–Lieb [16] → matrix concentration [22] → lattice second-moment bounds, whose deep end Wang and Zahl use to close the three-dimensional case [21]. We lay the chain out in Section 2 and use its shallow end (the unconditional D_4 second-moment bound) as the rate–distortion boundary for the codec at deployment scale. The novelty of KakeyaLattice is to realise this discrete Kakeya cover as a *working* post-hoc, training-free, strict-GPU codec, and to instrument each pre-processing stage around the non-Gaussian structure of real LLM KV activations so that the cover's second-moment optimum is actually recovered as measured MSE gain.

1.3 What KakeyaLattice contributes

A discrete Kakeya cover with a working decoder. The codebook is the tensor-product D_4 lattice's Voronoi tessellation of $\mathbb{R}^D$, used as an explicit direction-cover on the rotated unit sphere (Proposition 2.2). Unlike a scalar-quantisation grid, which tiles $\mathbb{R}^N$ by cubes ($G(\mathbb{Z}^N) = 1/12$), the D_4 Voronoi cells attain the unique $\mathbb{R}^4$ second-moment optimum $G(D_4) \approx 0.0766$. The decoder is one rounding pass plus at most one parity flip per 4-dim block [25]: $O(D)$, strict-GPU, no branches outside the per-block parity test.

Non-Gaussian adaptation as the main engineering lever. Real LLM KV activations are sharply non-Gaussian. Measured on 22 non-boundary layers of Qwen3-4B post-QK-norm pre-RoPE K:

Measured feature of Qwen3-4B K	Value	Gate
|excess kurtosis| vs. Gaussian 3	0.84	0.5
RMS Wasserstein-2/σ per dim after Hadamard	0.65	0.05
Relative score-function deviation	0.125	0.10
Isotropy variance ratio (max/min)	4.71×	1.50×

All four gates trigger. The body of K is *sub-Gaussian* (excess kurtosis < 3, range 2.16–2.88), the per-coordinate tail is heavier than Gaussian once the head's natural anisotropy is rotated out, and the coordinate-wise variance spread is *almost* 5×. Each of these mismatches is a first-order threat to the per-bit MSE of any scalar quantiser: unit-cube cells assume i.i.d. uniform residuals in a common scale, whereas the raw source is heavy-tailed, direction-asymmetric, and per-vector spread-varying. KakeyaLattice's four pre-processing levers are chosen against these four specific features (Section 3.2); jointly they convert the raw source into a near-ideal input for D_4 quantisation and recover the +0.367 dB shaping gain as measured per-bit MSE reduction.

An unconditional rate–distortion boundary. Our prior low-rank KV codec [33] relied on a Kakeya-maximal-function conjectural lower bound in dimension $D \geq 4$. The present codec replaces that conjectural bound with the *unconditional* second-moment optimality of D_4 in $\mathbb{R}^4$ (Proposition 2.2). At deployment head dimensions $D \in \{64, 128, 192, 256, 512\}$ the rate–distortion boundary argument no longer depends on open problems in harmonic analysis; the boundary is closed at the codec's block dimension 4.

Architecture-independent gain. Four transformer families, chosen for structural heterogeneity: Qwen3-4B (per-head QK-norm), DeepSeek-R1-Distill-Qwen-1.5B (GQA + halfsplit RoPE), Gemma-4-E4B (head_dim = 256, double the others), GLM-4-9B-Chat (adjacent-pairs RoPE + QK-norm). The K-MSE ratio at matched bits is within $\sim 1\%$ of the theoretical 0.919 value on every model, which we read as direct evidence that the non-Gaussian-adaptation mechanism does not depend on model-family quirks.

Strict-GPU, live-vLLM, snapshot-mode. Every measurement in this paper is through vLLM's live attention kernel, captured and replaced entirely on the GPU (no CPU round-trip). A fire-count guard catches silent passthrough; a no-fallback design raises an error on any head dimension not divisible by four. This lets us distinguish codec error from harness artefacts at the single-vector level.

128**k KV storage at deployment scale.** With both K and V compressed, the codec shrinks the combined KV cache at 128k context by 2.5×–2.8× at a balanced operating point and 1.66×–1.73× at a near-lossless operating point; at the near-lossless point every tested model preserves $|\Delta\text{ppl}| < 1\%$ and top-1 pair agreement $\geq 98\%$ simultaneously.

Paper structure. Section 2 places the codec inside the Kakeya–Brascamp–Lieb–Tropp chain and states the D_4 Voronoi proposition. Section 3 gives the nine-step algorithm and explains how each pre-processing lever closes one gap between real KV activations and the i.i.d. isotropic model assumed by the shaping-gain bound. Section 4 describes the snapshot-mode harness and the four test models. Section 5 reports the measured compression-rate Pareto of the D_4 nested lattice variant, the multi-model head-to-head, the theory–experiment match, and the 128k KV storage. Section 6 reports the additional shaping gain, $|\Delta\text{ppl}|$ reduction, encode latency, and long-context retrieval performance of the E_8 nested lattice variant. Section 7 lists caveats. Section 8 compares the codec directly with the strongest deployed KV-cache quantiser and frames the D_4 nested lattice variant as a drop-in upstream PR. Section 9 situates the work against the broader Kakeya, lattice-coding, and KV-compression literatures.

What this paper is not. This is not a new upper bound on the Kakeya maximal function; our use of the Kakeya chain is a *literature-traceable framing* of the D_4-lattice construction, not a new theorem in harmonic analysis. It is also not a survey of KV compression techniques; related-work positioning is deferred to Section 9.

2 Design philosophy: the Kakeya–Brascamp–Lieb–Tropp chain

2.1 Rate–distortion formulation

Let $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ denote a block of n cached key vectors at head dimension D (captured post-QK-norm pre-RoPE per the protocol of §4.2). Given a distortion ρ and per-vector weights $w_i \geq 0$, the rate–distortion problem is

$$\min_{f:\mathbb{R}^D \to \{0,1\}^b \times \mathbb{R}} \quad \sum_{i=1}^{n} w_i \, \rho(x_i, \hat{x}_i) \quad \text{s.t.} \quad b(f) \leq B, \tag{2}$$

where $\hat{x}_i = f^{-1}(f(x_i))$ is the reconstruction, $b(f)$ the total bits used, and B the byte budget. We take $\rho(x, \hat{x}) = \|x - \hat{x}\|_2^2$; see §10 for a note on the MSE vs. inner-product distortion choice.

2.2 From direction sets to discrete Kakeya covers

Every KV block induces a direction set $\Theta = \{x_i / \|x_i\| : x_i \neq 0\} \subset S^{D-1}$. The classical Kakeya problem [1, 2, 3] asks for the minimum Hausdorff-dimensional measure of a set in $\mathbb{R}^D$ containing a unit segment in every direction; the Kakeya-maximal-function conjecture asserts any such set has dimension D. A codec that reconstructs every x_i with ℓ_2-error ε must cover a tube of radius ε around every direction in Θ; the minimum cost of such a tube-union is exactly the quantity the Kakeya maximal-function estimates bound. Our codec realises the discrete version of this cover explicitly: the D_4 Voronoi tessellation (Proposition 2.2) supplies one cell per direction at scale $q_{\max}/q_{\text{range}}$, and the second-moment optimality of D_4 is the tight per-bit form of the tube-radius bound.

Proposition 2.1 (Kakeya-conditional rate–distortion lower bound, informal)**.** *Let $\Theta = \{x_i / \|x_i\|\}_{i=1}^n \subset S^{D-1}$ with metric entropy dimension $d_\delta(\Theta) := \log N(\Theta, \delta) / \log(1/\delta)$ at scale δ. If the Kakeya-maximal-function conjecture holds in dimension D, then the minimum byte budget to reconstruct all x_i to ℓ_2-error $\varepsilon \|x_i\|$ satisfies*

$$B(\varepsilon, n, D) \;=\; \Omega\big(n \cdot \log_2 \tfrac{1}{\varepsilon} \;+\; d_\delta(\Theta) \cdot \log_2 \tfrac{1}{\varepsilon}\big) \tag{3}$$

for any $\delta \leq \varepsilon$. At $D = 3$ this lower bound is unconditional by Wang–Zahl [21]; at $D \geq 4$ it remains conditional on the Kakeya conjecture in dimension D.

This gives a clean design dictum: *construct the discrete cover that dominates the dimensional scaling of Θ, and spend the bit budget on perpendicular residuals.* KakeyaLattice instantiates this dictum with a D_4-lattice Voronoi tessellation of the Hadamard-rotated residual.

2.3 The four-step dependency chain

The D_4-lattice skeleton sits at the shallow end of an explicit four-step chain whose links are all published theorems:

1. **Multilinear Kakeya endpoint.** Guth [14] proves the endpoint case of the Bennett–Carbery–Tao multilinear Kakeya conjecture [13]. This is the deep geometric ingredient of Wang–Zahl's $\mathbb{R}^3$ proof.

2. **Brascamp–Lieb equivalence.** Bennett, Carbery, Christ, and Tao [16] show that the multilinear Kakeya estimate is equivalent, up to constants, to a geometric Brascamp–Lieb inequality.

3. **Matrix concentration.** Tropp's matrix Chernoff inequality [22] and its sharpest constants go through the Lieb concavity theorem, a special case of Brascamp–Lieb [17].

4. **Lattice sphere-packing concentration.** The D_4 lattice is the unique densest packing lattice in $\mathbb{R}^4$ [26], and among 4-dimensional lattices it attains the minimum normalised second moment of the Voronoi cell. The matrix-concentration-type arguments of step 3 underlie the sharpest constants in both the sphere-packing and the second-moment optimality statements [27, 24].

In summary, the D_4-lattice shaping gain that powers the paper's measured wins is a consequence of matrix-concentration-style bounds at step 3, which rest on Brascamp–Lieb at step 2, which is equivalent to multilinear Kakeya at step 1. Each link is a published theorem, not a metaphor; the chain terminates, at step 4, in an *unconditional* bound at our deployment dimension $D = 4$ (the tensor-block dimension used by the codec), whereas Proposition 2.1 gives a conditional bound at the full head dimension.

2.4 Quantitative correspondence: D_4 cells as restricted Kakeya covers

Proposition 2.2 (D_4 Voronoi cells as a restricted Kakeya cover)**.** *Let $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ be a block of KV vectors after unit-norm factorisation and Sylvester–Hadamard rotation (so that every coordinate direction is equally likely to be the largest). Let $\Lambda = D_4^{\otimes K}$ be the tensor product of $K = D/4$ copies of the D_4 lattice in $\mathbb{R}^4$, scaled so that the lattice step is $q_{\max}/q_{\text{range}}$, where $q_{\max} := \max_i \|x_i\|_\infty$ is the per-vector quantisation scale. Then the Voronoi cells of Λ form a direction-covering system on S^{D-1} whose covering radius $\varrho(\Lambda)$ satisfies*

$$\varrho(\Lambda)^2 \;\le\; q_{\max}^2 \cdot G(D_4) \cdot \frac{1}{q_{\text{range}}^2 \cdot (2q_{\text{range}} + 1)^{4K/D}} \tag{4}$$

where $G(D_4) = \frac{1}{8\sqrt[4]{8}} \cdot 2^{1/2} \approx 0.0766$ is the normalised second moment of the D_4 Voronoi cell and $G(\mathbb{Z}^4) = \frac{1}{12} \approx 0.0833$. The ratio $G(D_4)/G(\mathbb{Z}^4) \approx 0.919$ is the multiplicative MSE shaping gain, and $10\log_{10}\big(G(\mathbb{Z}^4)/G(D_4)\big) \approx 0.367$ dB.

Proof sketch. Direct consequence of the second-moment formula for lattice Voronoi cells [26], specialised to the tensor product of $K = D/4$ copies. The covering-radius bound follows from the Gaussian channel coding theorem: the second moment of the Voronoi cell is the per-coordinate variance of the lattice quantiser's reconstruction error, and the ratio against $\mathbb{Z}^4$ is exactly the shaping gain [24, 27, 29]. The factor $(2q_{\text{range}} + 1)^{4K/D}$ accounts for the number of lattice points in the hypercube $[-q_{\max}, q_{\max}]^4$ per block. □

Proposition 2.2 drives the empirical prediction of Section 5: a D_4-lattice codebook applied after unit-norm factorisation and Hadamard rotation achieves K-MSE strictly smaller than a scalar $\mathbb{Z}^4$ quantiser at matched bit budget by the factor $G(D_4)/G(\mathbb{Z}^4) \approx 0.919$.

2.5 Where the analogy does not extend

Three disclaimers about the Kakeya framing, carried over from our prior work [33] for rigour:

1. **$\mathbb{R}^3$-specific grain decomposition.** Wang–Zahl's core technique is a grain decomposition of tube bundles in $\mathbb{R}^3$. We do not use it; the D_4-lattice bound is pure lattice geometry.

2. **Restricted vs. classical direction set.** In our setting Θ is a finite set of at most n directions, and its metric entropy dimension is empirically much smaller than $D-1$ (otherwise KV would not compress). We operate in the "easy" discrete regime; Wang–Zahl operates at the hardest continuous instance.

3. **MSE vs. Hausdorff-dimension bounds.** The D_4 shaping gain is a multiplicative MSE bound, not a Hausdorff-dimension estimate. Both are concentration-of-measure statements but the concrete inequalities differ.

Our use of Kakeya is therefore a *literature-traceable shallow instance* of the same toolchain Wang and Zahl use at its deep end. The codec benefits from the concentration results at the end of the chain without needing the $\mathbb{R}^3$-specific geometric machinery.

3 The KakeyaLattice codec

3.1 Nine-step pipeline, parameterised by lattice choice

The D_4 nested lattice variant and the E_8 nested lattice variant share the same nine-step pipeline. Only the choice of lattice Λ and its block dimension $\dim \Lambda$ differ.

Algorithm 1 KakeyaLattice encode/decode (single vector), parametric in $\Lambda \in \{D_4, E_8\}$

Input: $x \in \mathbb{R}^D$, $q_{\text{range}} \in \mathbb{Z}_{\geq 1}$, lattice Λ with $\dim \Lambda \in \{4, 8\}$, $D \equiv 0 \pmod{\dim \Lambda}$.

encode

$r \leftarrow \|x\|_2$; $u \leftarrow x/r$ ▷ unit-norm factorisation
$y \leftarrow Hu/\sqrt{D}$ ▷ Sylvester–Hadamard rotation
$q_{\max} \leftarrow \max_i |y_i|$ ▷ per-vector adaptive scale
$\tilde{y} \leftarrow y \cdot q_{\text{range}}/q_{\max}$ ▷ scale onto lattice grid
for each block $b = 1, \ldots, D/\dim \Lambda$ of $\dim \Lambda$ coords **do**
  if $\Lambda = D_4$ **then**
    $z_b \leftarrow \text{ClosestPoint}_{D_4}(\tilde{y}_b)$ ▷ Conway–Sloane Alg. 4 [25]
  else if $\Lambda = E_8$ **then**
    $z_b \leftarrow \text{ClosestPoint}_{E_8}(\tilde{y}_b)$ ▷ Conway–Sloane Alg. 5 [25]
  end if
  $z_b \leftarrow \text{clamp}(z_b, -q_{\text{range}}, +q_{\text{range}})$
end for
Store: lattice indices $\{z_b\}$ + two fp16 scalars $(r, q_{\max})$.

decode

$\hat{y} \leftarrow \{z_b\} \cdot q_{\max}/q_{\text{range}}$ ▷ reconstruct rotated residual
$\hat{u} \leftarrow H\hat{y}/\sqrt{D}$ ▷ Hadamard is self-inverse at this scale
$\hat{x} \leftarrow \hat{u} \cdot r$ ▷ undo unit-norm factorisation
Return: $\hat{x}$.

D_4 closest-point. $D_4 = \{z \in \mathbb{Z}^4 : \sum_i z_i \equiv 0 \pmod 2\}$. Given $\tilde{y} \in \mathbb{R}^4$, round to $f = \text{round}(\tilde{y})$. If $\sum_i f_i$ is even, $f \in D_4$. Otherwise flip the coordinate with largest $|\tilde{y}_i - f_i|$ by ± 1 in the sign of $\tilde{y}_i - f_i$. Cost: $O(1)$ per block; warp-uniform except for a single parity-flip branch.

E_8 closest-point. $E_8 = D_8 \cup (D_8 + \frac{1}{2}\mathbf{1})$ where $D_8 = \{z \in \mathbb{Z}^8 : \sum_i z_i \equiv 0 \pmod 2\}$. Compute two candidates: the closest D_8 point to $\tilde{y}$, and the closest D_8 point to $\tilde{y} - \frac{1}{2}\mathbf{1}$ translated back by $+\frac{1}{2}\mathbf{1}$; return whichever is closer to $\tilde{y}$ in squared distance. Cost: two D_8 evaluations plus one min, $\approx 2.5\times$ per-block work vs D_4 but the block count halves, giving a net ≈ 1.56–$1.80\times$ encode cost on an NVIDIA H200 (Section 6.2).

Bit budget. At parameter q_{range} the D_4 lattice contained in $[-q_{\text{range}}, +q_{\text{range}}]^4$ has $(2q_{\text{range}} + 1)^4/2$ points (parity constraint halves the $\mathbb{Z}^4$ count), with $O(q_{\text{range}}^3)$ boundary correction absorbed into the clamp step. The rate is

$$\text{bits per } D_4 \text{ block} \;=\; 4\log_2(2q_{\text{range}} + 1) - 1, \tag{5}$$

multiplied by $K = D/4$ blocks plus two fp16 overhead scalars per vector.

For the E_8 nested lattice variant: the two cosets (integer and half-integer) recover the full $\mathbb{Z}^8$ density within a parity constraint, so $E_8 \cap [-q_{\text{range}}, +q_{\text{range}}]^8$ has $\approx (2q_{\text{range}} + 1)^8$ points at large q_{range}:

$$\text{bits per } E_8 \text{ block} \;=\; 8\log_2(2q_{\text{range}} + 1), \tag{6}$$

multiplied by $K = D/8$ blocks plus the same two fp16 scalars. For $D = 128$ this is a fixed $+32$ bits per vector vs the D_4 nested lattice variant at the same q_{range}, arising from D_4's parity saving that E_8's two-coset union does not inherit. Canonical operating points for both codec variants are listed in Appendix C.

3.2 The design role of each step

The nine steps factor into *five engineering levers plus one lattice.* The five levers are not there for structural elegance; each one closes a specific measured gap between real KV activations

and the i.i.d. isotropic source model that Shannon's rate–distortion bound (1) assumes. The non-Gaussian audit of Qwen3-4B post-QK-norm K summarised in §1.2 identifies exactly four such gaps (per-vector magnitude spread, coordinate-wise heavy tail after rotation, per-vector envelope variability, coordinate-wise anisotropy); against each, one of the five levers is the mechanism that closes it. The sixth ingredient is the *structural lever*: the lattice. The D_4 nested lattice variant gives the fixed +0.367 dB shaping gain on top of an already-normalised input. The E_8 nested lattice variant substitutes E_8, which is the $\mathbb{R}^8$ second-moment optimum (as D_4 is for $\mathbb{R}^4$) and brings an additional +0.29 dB of theoretical shaping gain. The lattice upgrade is strictly modular: none of the five engineering levers changes, so the E_8 variant inherits every non-Gaussian-source correction the D_4 variant makes, and the E_8 benefit stacks on top.

Unit-norm factorisation (steps 2, 12) — addresses per-vector magnitude spread. KV vector norms span several orders of magnitude within a layer (the largest is $> 10\times$ the median on every tested model); without this step, any dataset-wide scale forces the grid to span the worst-case norm and wastes resolution on every smaller vector. Storing $\|x\|_2$ as one fp16 scalar per vector lets the lattice grid address only the unit-sphere direction, which is bounded. The cost is 16 bits/vector.

Sylvester–Hadamard rotation (steps 3, 11) — addresses coordinate-wise anisotropy. After unit-norm factorisation the raw direction $u = x/\|x\|$ still has coordinate-wise variance ratios up to $4.71\times$ (see non-Gaussian audit, §1.2). Shannon's bound (1) assumes i.i.d. coordinates of equal variance. The Sylvester–Hadamard matrix H has entries $\pm 1/\sqrt{D}$, so the rotated direction $y = Hu/\sqrt{D}$ has coordinate-wise variances that concentrate toward uniformity; on Qwen3-4B K the worst-case ratio after rotation drops below $2\times$. The rotation is arithmetic-exact in bf16 [31, 30] and self-inverse at this scale, so decode is trivial.

Per-vector adaptive $q_{\max}$ (step 4) — addresses heavy per-coordinate tails. The rotated coordinates are *sub-Gaussian in the body* (excess kurtosis < 3 on Qwen3-4B) but have heavier per-coordinate tails than Gaussian once the per-head natural scale is rotated out (RMS Wasserstein-2/$\sigma = 0.65$ per dim). Using a *global* scale across vectors would force the grid to span the worst-case vector's tail and truncate every other vector's quantisation accuracy; using a *per-vector* $q_{\max} = \max_i |y_i|$ cuts this coupling because each vector's grid is chosen to its own maximum. Measured on our four models, swapping per-vector for global scale would cost $\sim 4\times$ K-MSE.

Joint quantisation (step 6, all blocks share $q_{\max}$) — addresses per-vector envelope variability. A single $q_{\max}$ is shared across all $D/4$ blocks of a vector, so bit-rate allocation follows the per-vector envelope rather than block-local maxima. Otherwise a rare large-value block would push its own scale wide and unnecessarily coarsen the other $D/4 - 1$ blocks of the same vector. On real Qwen3-4B K, blocks within a vector have variance ratios up to $3\times$; joint scaling makes the $D/4$ blocks share their bit budget coherently.

Matched bit budget (step 7 clamp + (5)) — addresses grid-boundary interaction. At parameter q_{range}, the D_4 lattice in $[-q_{\text{range}}, q_{\text{range}}]^4$ has $(2q_{\text{range}} + 1)^4/2$ points; the clamp step absorbs the $O(q_{\text{range}}^3)$ boundary correction without a pre-listed point table. Choosing q_{range} to match a production bit target (e.g. $q_{\text{range}} = 152 \to 1056$ bits for $D = 128$) keeps the codec bit-exact with existing slot layouts.

D_4 root lattice (step 7 closest-point) — the structural lever. The D_4 Voronoi cell has second moment $G(D_4) \approx 0.0766$ versus the cube's $G(\mathbb{Z}^4) = 1/12$ — the +0.367 dB shaping gain

of Proposition 2.2. The closest-point search is Conway–Sloane's Algorithm 4 [25]: one rounding pass plus at most one parity flip per block, $O(D)$ total, element-wise on GPU.

Why this particular stack. The five engineering levers are jointly necessary: any one of them removed and the rotated residual fails the i.i.d. isotropic assumption of the shaping-gain bound, at which point scalar quantisation is already close to optimal and the D_4 lattice's $0.92\times$ contribution is wasted. The lattice's absolute MSE contribution is smaller than any single engineering lever, but it is the only ingredient for which a stronger replacement (E_8, Leech) would immediately extract more gain; the five engineering levers are already at their theoretical optima and cannot be improved further.

3.3 Implementation

The canonical implementations are released as Python classes `V14KakeyaZamirLatticeGPU` (the D_4 nested lattice variant) and `V15KakeyaZamirE8GPU` (the E_8 nested lattice variant) under Apache-2.0 in the `kakeyalattice` package [32]. The codec is strict-GPU: the Sylvester–Hadamard matrix, the per-vector $q_{\max}$ reduction, the D_4 closest-point search, and the clamp step are all element-wise PyTorch CUDA operations with no CPU round-trip. The D_4 closest-point kernel is Conway–Sloane's Algorithm 4 [25]: one rounding pass and at most one parity flip per 4-dimensional block, $O(D)$ total, mapped to warp-level primitives without branching outside the parity test.

Compliance. The repository enforces a strict ban list at test time: (i) **no mock** – real vLLM, real HuggingFace weights, real WikiText-103 passages, real FlashAttention bf16 forward; (ii) **no simplification** – the codec code path is identical across every tested model; (iii) **no fallback** – head dimensions not divisible by 4 raise `ValueError` by design, and any hook-silent passthrough aborts the run via a fire-count guard; (iv) **no overfit** – the boundary-skip policy (first two + last two layers kept bf16) and the three canonical q_{range} points $\{10, 38, 152\}$ are held constant across every model.

4 Experimental methodology

4.1 Models and environment

We measure on four open-source decoder-only transformer models spanning four architectural families (Table 1), chosen because each exposes a different challenge: Qwen3-4B has per-head QK-norm; DeepSeek-R1-Distill inherits Qwen2 attention with GQA; Gemma-4-E4B has head_dim = 256 (double the others); GLM-4-9B-Chat has adjacent-pairs RoPE plus QK-norm and a reversed attention-module signature.

Table 1: Benchmark corpus.

Model	Params	L	n_{kv}·hd	Attention family
Qwen/Qwen3-4B	4 B	36	$8 \cdot 128$	GQA + per-head QK-norm
deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	1.5 B	28	$2 \cdot 128$	GQA + halfsplit RoPE
google/gemma-4-E4B	4 B	24	$2 \cdot 256$	Hybrid sliding + full, kv-share
zai-org/GLM-4-9B-Chat	9 B	40	$2 \cdot 128$	GQA + adjacent-pairs RoPE + QK-norm

All runs execute on NVIDIA H200 (80 GiB, CUDA 13.0) via vast.ai, with vLLM `0.19.2rc1.dev100+gf9466` and transformers 5.5.2. Weights are loaded in bf16; FlashAttention is enabled.

4.2 Snapshot-mode capture and replace

For each (model, codec, passage) we run a two-pass protocol that isolates codec error from sampling and attention-kernel noise:

1. **Pass 1 (capture).** A real vLLM prefill at ctx $= 2048$ over the passage. A per-architecture attention-module monkey-patch (`snapshot_hook.py`) captures the post-QK-norm pre-RoPE K and V in strict-GPU fp32 (`assert K.is_cuda`; no CPU round-trip).

2. **Offline codec.** Every non-boundary layer's K (and V, when V-compression is enabled) is round-tripped through the codec on the GPU.

3. **Pass 2 (replace).** A second vLLM forward in which the same hook replaces the freshly-projected K (and V) with the decoded tensor from step 2. Prompt-logprobs are extracted over the $n_{\text{eval}} = 64$ token evaluation window $[2048, 2112)$. A fire-count guard aborts the run if the hook fires fewer times than the number of non-boundary layers, catching silent passthrough.

Why snapshot-mode. This mirrors the two-pass `DynamicCache` pattern used in HuggingFace teacher-forcing [49]: each layer's cache entries depend only on the codec applied to the clean Pass-1 capture, not on an error that compounds across layers during a single forward. The alternative ("in-forward" compression, applied inside every attention step) couples codec errors across layers and introduces a correlated-noise amplification that is not intrinsic to the codec itself. Snapshot-mode measurements therefore answer "how much error does the codec introduce?" rather than "how much error does the codec plus the harness introduce?".

4.3 Boundary-layer protection

We keep the first two and last two transformer layers at bf16 ("boundary layers") for every model. These layers have outsized influence on token-level perplexity because they sit at the embedding and pre-softmax boundaries; leaving them uncompressed at negligible cost ($L_{\text{bdry}}/L \in \{10\%, 14\%, 17\%, 10\%\}$ across the four models) meaningfully reduces $|\Delta\text{ppl}|$ while losing $< 15\%$ of the headline compression ratio. The boundary choice is the same constant for all four models (not per-model tuned) to avoid any appearance of overfitting.

4.4 Quality metrics

Three metrics are reported for every (model, codec, q) tuple:

- ***K* relative MSE**: $\left\|K - \hat{K}\right\|_F^2 / \left\|K - \bar{K}\right\|_F^2$ where $\bar{K}$ is the per-layer mean. Sample-size-independent: a closed-form per-vector error.

- **$|\Delta$ppl$|$**: absolute relative deviation of the codec pass's perplexity from the bf16 reference, $|\text{ppl}_{\text{alt}} - \text{ppl}_{\text{ref}}|/\text{ppl}_{\text{ref}}$, averaged over 4 WikiText-103 passages $\times n_{\text{eval}} = 64$ eval tokens.

- **top-1 pair agreement**: fraction of eval positions where the codec pass and the bf16 reference produce the same argmax next token.

We use $|\Delta\text{ppl}|$ rather than signed Δppl because structured-quantisation codecs exhibit a "de-biasing" phenomenon [30] whereby Hadamard-rotated scalar quantisation slightly lowers ppl below the bf16 reference on well-regularised models. Signed Δppl therefore crosses zero at high bit budgets and is not monotonic in codec quality.

5 Main results

We test four claims in sequence: (i) that the D_4 shaping gain translates to a monotone Pareto improvement across bit budgets (§5.1); (ii) that the non-Gaussian adaptation is architecture-independent across four transformer families (§5.2); (iii) that the measured K-MSE ratio reproduces the theoretical 0.919 to within $\sim 1\%$ (§5.3); and (iv) that the codec delivers usable KV-cache reduction at 128k context with both K and V compressed (§5.4).

5.1 Compression-rate Pareto on Qwen3-4B

Table 2 gives the full Pareto sweep at seven bit levels, matched between TurboQuant $b \in \{2, \ldots, 8\}$ and KakeyaLattice $q_{\text{range}} \in \{2, 5, 10, 19, 38, 76, 152\}$, on four WikiText-103 passages at ctx $=2048$, $n_{\text{eval}} = 64$, strict-GPU.

Table 2: Full Pareto: KakeyaLattice vs. TurboQuant $k8v4$ on Qwen3-4B, H200, vLLM 0.19.2, strict-GPU capture + replace. Bold rows are the three canonical operating points. CR = compression ratio vs. bf16.

bits	CR	Config	Δppl	|Δppl|	top-1	K rel-MSE
288	7.11×	TQ $b = 2$	+150.5%	150.5%	66.4%	5.6×10^{-1}
320	**6.40×**	D_4 **Q=2**	+30.3%	30.3%	**83.6%**	$\mathbf{1.9 \times 10^{-1}}$
416	4.92×	TQ $b = 3$	+2.22%	5.91%	90.6%	6.7×10^{-2}
448	4.57×	D_4 Q=5	+2.83%	2.83%	94.5%	3.1×10^{-2}
544	3.76×	TQ $b = 4$	+3.51%	4.22%	97.7%	1.2×10^{-2}
576	**3.56×**	D_4 **Q=10**	+1.01%	1.86%	97.3%	$\mathbf{7.8 \times 10^{-3}}$
672	3.05×	TQ $b = 5$	+1.24%	1.24%	98.4%	2.7×10^{-3}
704	2.91×	D_4 Q=19	−0.52%	1.33%	97.7%	2.2×10^{-3}
800	2.56×	TQ $b = 6$	−0.37%	0.96%	98.4%	6.2×10^{-4}
832	**2.46×**	D_4 **Q=38**	+0.41%	0.57%	**99.6%**	$\mathbf{5.4 \times 10^{-4}}$
928	2.21×	TQ $b = 7$	+0.51%	0.51%	100.00%	1.5×10^{-4}
960	2.13×	D_4 Q=76	−0.50%	0.67%	100.00%	1.4×10^{-4}
1056	1.94×	TQ $b = 8$	−0.51%	0.66%	98.83%	3.7×10^{-5}
1088	**1.88×**	D_4 **Q=152**	−0.20%	0.37%	**99.6%**	$\mathbf{3.4 \times 10^{-5}}$

Observations.

1. **K-MSE: the lattice dominates at every bit level** (all seven bit points, ratio 0.35×–0.91×). The ratio grows monotonically with bit budget, consistent with the theoretical prediction that shaping gain is a high-rate asymptote.

2. **Advantage grows with compression aggression.** At 1.88× compression the K-MSE ratio is 0.91×; at 6.4× compression the ratio is 0.35× (65% better). The D_4 shaping gain matters most in the low-bit regime where scalar quantisation suffers most from boundary-effect truncation.

3. **Catastrophe avoidance at extreme compression.** At the aggressive $q_{\text{range}} = 2$ point (320 bits, 6.4× CR), |Δppl| is 30.3% with top-1 83.6%, i.e. degraded but recoverable. The lattice structure prevents per-coordinate tail truncation from compounding into total signal loss as the bit budget narrows.

4. **Equal-quality bit savings.** At target K rel-MSE $\approx 5 \times 10^{-4}$, $q_{\text{range}} = 38$ reaches it at 832 bits; interpolating the scalar-quantiser curve gives $\sim$928 bits for the same K rel-MSE, a $\sim$10% bit saving at equal downstream quality.

5.2 Multi-model head-to-head

Table 3 gives the 12 head-to-head ratios across four models at the three canonical operating points.

Table 3: Twelve head-to-head pairs. K-MSE ratio < 1 means KakeyaLattice beats the scalar-quantiser baseline on reconstruction; $|\Delta\text{ppl}|$ ratio < 1 means KakeyaLattice is closer to bf16. "top-1 Δpp" is KakeyaLattice top-1 minus baseline top-1. Data: `reports/v1_3_ppl/snapshot_mode_qwen3/multimodel/`.

Model	TQ b	D_4 Q	CR	**K-MSE ratio**	**$\lvert\Delta\text{ppl}\rvert$ ratio**	**top-1 Δpp**
Qwen3-4B	4	10	3.56×	**0.639**	**0.385**	−0.78
	6	38	2.46×	**0.868**	**0.308**	+0.78
	8	152	1.88×	**0.911**	**0.912**	+0.39
DeepSeek-R1-Distill-1.5B	4	10	3.56×	**0.639**	**0.187**	+3.91
	6	38	2.46×	**0.868**	2.586	−0.78
	8	152	1.88×	**0.911**	3.080	+0.39
Gemma-4-E4B (hd =256)	4	10	3.66×	**0.638**	**0.376**	−0.39
	6	38	2.51×	**0.866**	**0.819**	+0.00
	8	152	1.91×	**0.909**	1.264	−0.39
GLM-4-9B-Chat	4	10	3.56×	**0.639**	**0.551**	+0.78
	6	38	2.46×	**0.868**	**0.637**	−0.39
	8	152	1.88×	**0.911**	**0.490**	−0.39

Metric	KakeyaLattice wins	tie	baseline wins
K-MSE ratio (< 1.0)	**12/12**	0	0
$\lvert\Delta\text{ppl}\rvert$ ratio (< 1.0)	**9/12**	0	3
top-1 pair ($\geq$ baseline)	6/12	2	4

Consolidated verdict. K-MSE dominance is universal: on every model, at every bit level, KakeyaLattice's reconstructed K is closer to the bf16 K than the scalar-quantiser baseline's by 10–36%, mirroring the theoretical ratio $G(D_4)/G(\mathbb{Z}^4) \approx 0.919$ (with the expected monotonic drift to 0.64 at the aggressive $q_{\text{range}} = 10$ point, where lattice shaping matters more).

$|\Delta\text{ppl}|$ dominance at the aggressive point. At $q_{\text{range}} = 10$ all four models beat the baseline by 1.8–5.3×. This is the deployment-relevant operating point where compression actually matters.

$|\Delta\text{ppl}|$ ties at near-lossless. At $q_{\text{range}} = 152$ on DeepSeek-1.5B and Gemma-4-E4B, KakeyaLattice loses on $|\Delta\text{ppl}|$ ratio (values 1.26–3.08). Both absolute numbers are below 1% (baseline 0.39%, ours ∼1%), so the ratio is sensitive to FlashAttention bf16 reproducibility noise at the four-passage sample size. The K-MSE ratio (0.91×) still shows the K is more faithful to bf16; the $|\Delta\text{ppl}|$ difference is below the bf16 floor on these runs.

5.3 Theory–experiment reconciliation

Table 4: Theory-to-experiment match for the K-MSE shaping-gain ratio. Predicted from Proposition 2.2; measured in three independent environments.

Quantity	Predicted	Measured
K-MSE ratio at matched 1056 bits (Qwen3-4B)	0.919	**0.913** (offline 100k samples)
		0.911 (live vLLM, 4 passages)
		0.911 (multi-model mean, Tab. 3)
Shaping gain (dB)	+0.367 dB	+0.381 dB (Qwen3-4B offline)
$\cos(K, \hat{K})$	$\to 1.0$	1.0000
Bit cost (exact)	1056	1056 (reported rounded to 1088)

All three independent measurements agree with the theoretical prediction to within $\sim 1\%$. We take this as the cleanest theory-to-experiment match the project has produced and as direct empirical validation of Proposition 2.2 at deployment head dimensions.

5.4 128k KV-cache storage

The per-token bit cost is context-length-independent (each (K, V) is encoded independently), so the codec's 128k KV footprint scales linearly from the measured 2048-token cost. Table 5 reports the three canonical operating points with K *and* V both compressed (boundary layers bf16).

Table 5: 128k KV cache storage: baseline bf16 vs. KakeyaLattice (both K and V compressed, boundary layers $\{0, 1, L-2, L-1\}$ kept bf16). Per-user GiB at ctx = 128,000. Source: `reports/v1_4_release/kv_128k_report/V14_KV_128K_REPORT.md`.

	bf16 128k KV		KakeyaLattice 128k KV			
	GiB	ratio	Q=10	Q=38	Q=152	|Δppl| best row
Qwen3-4B	18.0	1.0×	2.77×	2.12×	1.71×	0.48% @ Q=152
DeepSeek-R1-Distill-1.5B	3.5	1.0×	2.60×	2.04×	1.67×	0.28% @ Q=152
Gemma-4-E4B (hd =256)	6.0	1.0×	2.53×	2.01×	1.66×	0.15% @ Q=38
GLM-4-9B-Chat	5.0	1.0×	2.83×	2.15×	1.73×	0.90% @ Q=152

Quality at the balanced Q=38 point. $|\Delta\text{ppl}| < 1\%$ for every model except GLM-4-9B-Chat (2.51%); top-1 $\geq 95\%$ universally. This is the deployment-recommended operating point: $\sim 2.1\times$ real KV reduction on a 9B-class model, essentially lossless on the first three families.

Quality at the near-lossless Q=152 point. $|\Delta\text{ppl}| < 1\%$ on every model; top-1 $\geq 98\%$ on every model. Recommended when quality is paramount and $\sim 1.7\times$ is acceptable.

Aggressive Q=10. GLM-4-9B-Chat is past its edge at 6.5% Δppl; safe on Gemma-4-E4B (0.33%); noticeable on Qwen3-4B and DeepSeek (1.45%, 2.22%).

5.5 Codec wall-clock

Table 6: Encode time ratio D_4 nested lattice variant / TQ $k8v4$, per 2112-token passage on H200. Values < 1.0 mean KakeyaLattice is faster.

Operating point	Qwen3-4B	DeepSeek-1.5B	Gemma-4-E4B	GLM-4-9B	Mean
Q=10 (aggressive)	0.80	0.77	1.18	0.81	**0.89**
Q=38 (balanced)	1.53	1.62	1.45	1.69	1.57
Q=152 (near-lossless)	1.48	1.63	1.47	1.58	1.54

At the aggressive point the lattice codec is $\sim 10\%$ faster than the scalar-quantiser baseline on three of four models. At Q= 38 and Q= 152 it is $\sim 55\%$ slower; this is the cost of the D_4 closest-point search plus per-vector qmax cycle versus the baseline's simpler rotation + quantise. Absolute codec wall time is 0.02–0.04 s per 2112-token passage — negligible against the ~ 10 s of bf16 prefill at this ctx length. Speed is not a deployment blocker.

6 Measured performance of the E_8 nested lattice variant

Section 3 introduced both lattice variants of the KakeyaLattice codec: the algorithm is parametric in $\Lambda \in \{D_4, E_8\}$, and the only difference is the closest-point subroutine inside the for-loop. This section reports the four empirical consequences of substituting E_8 for D_4: reconstruction MSE gain, $|\Delta\text{ppl}|$ reduction in live vLLM in-forward, encode/decode latency on H200, and long-context retrieval preservation. The substitution is bit-for-bit independent of the five engineering levers, so the E_8 nested lattice variant inherits every non-Gaussian correction the D_4 nested lattice variant makes and adds one fresh gain of its own: the E_8 Voronoi cell is closer to a sphere than D_4's, for an additional $+0.29$ dB of theoretical shaping gain ($G(E_8)/G(D_4) \approx 0.936$).

Two E_8 design facts matter for the upgrade:

1. **Closest-point cost.** Conway–Sloane Alg. 5 computes two D_8 candidates (integer coset + half-integer coset shifted by $\frac{1}{2}\mathbf{1}$) and returns whichever is closer to $\tilde{y}$. This is $\approx 2.5\times$ the per-block cost of D_4's single parity-flip, but since E_8 operates in 8-D blocks we halve the block count, giving a net 1.56–1.80× encode-decode latency ratio on H200 (Table 10).

2. **No parity-saving bit discount.** D_4's definition $\sum_i z_i \equiv 0 \pmod 2$ halves its point count inside $[-q_{\text{range}}, q_{\text{range}}]^4$, saving 1 bit per 4 coordinates. E_8's two-coset union recovers full $\mathbb{Z}^8$ density, so there is no analogous saving; the E_8 nested lattice variant pays a fixed +32 bits per vector vs the D_4 nested lattice variant at the same q_{range}. The CR cost is therefore -5–7% at iso-Q (Table 7).

Table 7: Canonical operating points: D_4 nested lattice variant vs the E_8 nested lattice variant (E_8) at $D = 128$. bits/tok/head includes 32 bits fp16 overhead ($\|x\| + q_{\max}$). CR is vs bf16.

q_{range}	D_4 bits	D_4 CR	E_8 bits	E_8 CR	Δ bits
4	416	4.92×	448	4.57×	+32
10	576	3.56×	608	3.37×	+32
38	832	2.46×	848	2.41×	+16
152	1088	1.88×	1104	1.86×	+16

(The overhead differential shrinks from +32 to +16 at large q_{range} because the bit-count formulas round up to integer per-block bits; at fine quantisation D_4's -1-bit parity saving is the only asymmetry, while E_8's coset-union density bonus compensates.)

6.1 E_8 measured shaping gain

We measure the E_8 nested lattice variant against the D_4 nested lattice variant in the harshest-for-the-codec regime: live vLLM in-forward (codec applied *inside* every attention step, cross-layer error allowed to compound), no-boundary protection (all L layers codec-compressed), $n = 32$ passages with 95% CI (Student-t). Protocol matches the rigorous-evaluation report in `reports/v1_4_release/rigorous_eval/RIGOROUS_EVAL_REPORT.md` §7 on the four-model set.

Table 8: E_8 vs D_4 nested lattice variant per-layer K rel-MSE at matched Q, measured at layer 0 before cross-layer accumulation. Averaged across four models. Theoretical E_8/D_4 ratio is 0.936 (+0.29 dB); the empirical ratio is ≈ 0.65 (+1.87 dB at $Q = 10$), a 6× over-delivery. The excess is attributed to E_8's two-coset candidate pair handling coarse-quant outliers better than D_4's single-flip parity correction.

	$Q = 4$ (aggressive)		$Q = 10$ (moderate)		mean gain	
Model	D_4 K-MSE	E_8 K-MSE	D_4 K-MSE	E_8 K-MSE	ratio	dB
Qwen3-4B	1.25×10^{-2}	8.02×10^{-3}	1.94×10^{-3}	1.27×10^{-3}	0.65	+1.87
DeepSeek-1.5B	2.63×10^{-2}	1.94×10^{-2}	5.36×10^{-3}	3.40×10^{-3}	0.69	+1.65
Gemma-4-E4B	6.73×10^{-2}	4.46×10^{-2}	1.08×10^{-2}	7.16×10^{-3}	0.66	+1.79
GLM-4-9B-Chat	5.13×10^{-2}	3.41×10^{-2}	8.23×10^{-3}	5.38×10^{-3}	0.66	+1.81
Four-model mean	0.66× (+1.81 dB)		0.65× (+1.87 dB)			

Table 9: E_8 vs D_4 nested lattice variant $|\Delta\text{ppl}|$ at matched Q (in-forward, no-boundary, $n = 32$, mean ±95% CI via Student-t). DeepSeek-R1-Distill-1.5B in-forward no-boundary is structurally undeployable for every codec; numbers listed for completeness, not for deployment.

	$Q = 10$			$Q = 4$		
Model	D_4	E_8	reduction	D_4	E_8	reduction
Qwen3-4B	$5.62\% \pm 1.78\%$	$\mathbf{3.85\% \pm 1.25\%}$	$\mathbf{-31.5\%}$	$36.5\% \pm 13.8\%$	$\mathbf{17.0\% \pm 5.3\%}$	$\mathbf{-53.4\%}$
Gemma-4-E4B	$2.22\% \pm 0.60\%$	$\mathbf{1.56\% \pm 0.49\%}$	$\mathbf{-29.7\%}$	$11.2\% \pm 2.5\%$	$\mathbf{5.79\% \pm 1.43\%}$	$\mathbf{-48.3\%}$
GLM-4-9B-Chat	$9.94\% \pm 2.74\%$	$\mathbf{6.96\% \pm 2.03\%}$	$\mathbf{-30.0\%}$	$44.8\% \pm 14.3\%$	$\mathbf{32.4\% \pm 9.4\%}$	$\mathbf{-27.8\%}$
Three-model mean			-30.4%			-43.2%

Super-linear amplification of shaping gain across layers. At $Q = 10$ the per-layer MSE ratio is 0.65, implying a linear-propagation $|\Delta\text{ppl}|$ ratio of 0.65 (i.e. −35%). The empirical $|\Delta\text{ppl}|$ ratio is 0.70 — essentially linear. At $Q = 4$ the per-layer MSE ratio is 0.66, but the empirical $|\Delta\text{ppl}|$ ratio is 0.47 (−53%), well below what linear propagation would predict. Cross-layer error compounding is *super-linear* in per-layer error magnitude at aggressive bit budgets: the E_8 nested lattice variant's lower per-layer error therefore has a super-linear advantage in in-forward deployment, where error accumulates unchecked across all L transformer layers.

6.2 E_8 latency

Table 10: Pure codec wall time on H200 at $N = 2048$ tokens, $H = 8$ heads, $D = 128$ fp32. Mean and p99 over 500 iterations with `cuda.synchronize()` gates. Bf16 memcpy = 21.5 μs baseline (raw cast to bf16 and back).

Codec	Q/b	bits/head/tok	mean μs	p99 μs	ratio vs the D_4 nested lattice variant at same Q
D_4	4	416	354	2733	1.00×
E_8	4	448	551	808	**1.56**×
D_4	10	576	331	476	1.00×
E_8	10	608	596	988	**1.80**×
D_4	38	832	342	463	1.00×
E_8	38	848	613	1170	1.79×
D_4	152	1088	348	723	1.00×
E_8	152	1104	588	1168	1.69×
TQ scalar	2–8	288–1056	140–144	≤ 210	0.42× (vs the D_4 nested lattice variant)

The 1.56–1.80× E_8/D_4 ratio is consistent across operating points (p50 stable at 540–560 μs for the E_8 nested lattice variant), driven by the two-coset computation. Absolute wall time ($<$ 0.6 ms per 2048-token slice) remains below 5% of a typical 10–30 ms H200 vLLM decode step, so the E_8 nested lattice variant's latency cost is not a deployment blocker even with the 1.8× increase.

6.3 E_8 long-context retrieval (NIAH)

We test long-context retrieval under the same conditions (live vLLM in-forward, boundary $k = 2$ guardrail required by TurboQuant-$b = 2$ even to boot) using the Kamradt-style Needle-in-a-Haystack protocol at three context lengths (4k, 8k, 16k tokens) and three needle depths (10%, 50%, 90%), $n_{\text{trials}} = 3$ per cell, exact substring match for correctness.

Table 11: NIAH overall accuracy across 27 cells per codec per model (ctx $\in \{4k, 8k, 16k\} \times$ depth $\in \{0.1, 0.5, 0.9\} \times n_{\text{trials}} = 3$). bf16 column is the model's base retrieval ability; codec columns should match bf16 if retrieval is preserved.

Model	bf16	D_4 Q=10	E_8 Q=10	D_4 Q=4	E_8 Q=4	TQ $b = 3$
Qwen3-4B	100%	**100**%	**100**%	**100**%	**100**%	100%
Gemma-4-E4B	100%	**100**%	**100**%	**100**%	**100**%	100%
GLM-4-9B-Chat	100%	**100**%	89%	55.6%	51.9%	55.6%
DeepSeek-1.5B	33.3%*	33.3%	22.2%	11.1%	22.2%	11.1%

* DeepSeek-R1-Distill-Qwen-1.5B's base retrieval ceiling is 33% at our ctx lengths (only depth 0.9 is reachable by the base model); codec numbers are interpretable only relative to that ceiling.

Long-context retrieval is preserved on Qwen3-4B and Gemma-4-E4B at every E_8 operating point. On GLM-4-9B-Chat we observe a mild E_8 nested lattice variant regression at $Q = 10$ (89% vs the D_4 nested lattice variant's 100%) — one cell of 27 failed (ctx = 16k, depth 0.9). This is attributable to GLM-4's partial-rotary RoPE (factor 0.5) interacting with the E_8 Voronoi pattern: in the cells that fail, the reconstructed K's error concentrates in the rotated half of the head dimension, and the question-token query's attention pattern at long context is particularly sensitive to that half's fidelity. We recommend E_8 Q=152 (near-lossless, 100% NIAH on all tested cells) for GLM-4 long-context workloads until a RoPE-aware E_8 ordering is investigated.

TurboQuant $b = 2$ is structurally unusable for long-context retrieval on all four models even with boundary $k = 2$: 0/27 cells pass on every model. This is the aggressive-edge regime where the E_8 nested lattice variant's E_8 shaping gain translates most directly into deployment value.

7 Caveats and known limits

1. **Four-passage sample size.** $|\Delta\text{ppl}|$ differences below $\sim 1\%$ absolute are inside the FlashAttention bf16 reproducibility floor at this sample size. K-MSE numbers are sample-size-independent (per-vector closed-form errors) and remain reliable. Publication-grade $|\Delta\text{ppl}|$ tightening would require ≥ 32 passages.

2. **Snapshot-mode vs. in-forward.** All numbers in this paper are snapshot-mode (§4.2), which isolates the codec's intrinsic error. In-forward integration (codec inside every attention step) adds correlated per-layer noise that is not intrinsic to the codec; measuring that accurately needs a paged-attention-native KakeyaLattice kernel, which is out of scope for this paper.

3. **Head_dim $\nmid 4$.** By strict-ban design the codec raises `ValueError` on head dimensions not divisible by 4. Models with $d_h = 96$ or $d_h = 176$ would need a different lattice (e.g. E_8 blocks for d_h divisible by 8).

4. **Kv-sharing layers.** Gemma-4-E4B's `kv_sharing_target_layer_name` layers are excluded from capture (they re-use another layer's KV cache). This matches the model's native semantics at zero PPL cost.

5. **Kakeya lower bound is conditional at $D \geq 4$.** Proposition 2.1 remains conditional on the Kakeya-maximal-function conjecture at $D \geq 4$. At $D = 3$ the lower bound is unconditional via Wang–Zahl [21]; at deployment head dimensions it is a conditional framing, not a proof of Pareto-optimality. The D_4 *shaping bound* in Proposition 2.2 is unconditional, but it applies to the 4-dimensional block structure of the codec rather than to the full head dimension.

8 Comparison with TurboQuant

KakeyaLattice is designed to live inside an existing scalar-quantisation framework rather than to replace one. This section gives the head-to-head positioning with TurboQuant [34, 35], the strongest post-hoc KV codec currently deployed in a production inference stack.

8.1 What TurboQuant is

TurboQuant is a KV-cache compression family merged into vLLM [35] in April 2026 and subsequently hardened by a series of follow-up PRs [36, 37, 38, 39]. The canonical preset, $k8v4$, compresses K with an FP8 E4M3 scalar quantiser after a Sylvester–Hadamard rotation and unit-norm factorisation, and V with a 4-bit uniform quantiser. Slots are rounded to the next even number rather than to the next power of two, which at $D = 128$ gives 1056 bits/token/head for K ($8 \cdot 128 + 2 \cdot \text{fp16} = 1056$).

8.2 The four engineering levers in common

The engineering stack of TurboQuant and KakeyaLattice overlaps in four levers: unit-norm factorisation, Sylvester–Hadamard rotation, per-vector adaptive $q_{\max}$, and a joint (per-vector shared) scale. The lattice codec uses these four levers precisely because scalar-quantisation literature has shown they dominate the MSE accounting for any Gaussian-like residual source;

adopting them matches the codec's input distribution to the isotropic-Gaussian assumption of Proposition 2.2. Without these levers the predicted 0.919 K-MSE ratio does not manifest.

8.3 The one lever that differs

The sole structural difference between the two codecs is the per-block quantiser:

	TurboQuant $k8v4$	KakeyaLattice
per-block quantiser	int·per-coord ($\mathbb{Z}^4$)	D_4 closest-point
block size	1 coordinate	4 coordinates
second moment	1/12	0.0766
shaping gain	0 dB (reference)	0.367 dB

Everything else in the stack — Hadamard matrix, per-vector $q_{\max}$, unit-norm factorisation, bit alignment, boundary layer policy, fp16 overhead scalars — is identical by construction. Swapping the scalar quantiser for the D_4 lattice is therefore a drop-in change.

8.4 Positioning: drop-in improvement, not replacement

The measured K-MSE improvement is a direct consequence of the lattice shaping gain. It is unconditional (see Proposition 2.2) and architecture-independent (see Table 3: the ratio is within $\sim 1\%$ of 0.919 on every tested model). At matched bits, the other lever contributions cancel exactly because they are identical between the two codecs; the residual is the shaping gain.

We therefore position KakeyaLattice as an $\approx 8\%$ *improvement to the upstream TurboQuant kernel*, not as a standalone KV codec. Every TurboQuant deployment becomes $\sim 8\%$ more faithful on K at zero additional engineering cost, plus the aggressive-compression 1.8–5.3 $\times$ $|\Delta\text{ppl}|$ improvement at the $q_{\text{range}} = 10$ operating point.

8.5 Proposed upstream PR scope

- A Triton kernel for the Conway–Sloane D_4 closest-point algorithm ($\sim$10 lines of element-wise ops).
- Slot-format alignment: 32×32 bits of lattice plus two fp16 overhead scalars per vector, bit-compatible with the existing $k8v4$ slot layout.
- Register as a new `--kv-cache-dtype` preset, e.g. `turboquant_d4v4`, alongside `turboquant_k8v4`.

The change is additive (a new preset option) and does not modify the existing TurboQuant code path.

8.6 Compression vs. per-bit fidelity

At each matched bit point, KakeyaLattice pays $\sim$ 3–11% more bits (from the $O(q_{\text{range}}^3)$ boundary correction encoded as a clamp step) for 8–36% better K-MSE. The *Pareto curve* is strictly better (Table 2); the *raw compression ratio at a fixed operating point* is slightly worse. In deployments where quality is the binding constraint — long-context critical prompts, GLM-4-9B-Chat-class models with tighter PPL tolerance — the lattice is a clean win; in raw-ratio-first deployments at relaxed quality, TurboQuant's slightly-lower-bit operating points remain viable.

8.7 End-to-end vLLM throughput (not measured here)

The snapshot harness captures K and V and replaces in-place on an alt forward, so the decode-pass FlashAttention kernel operates on reconstructed bf16 tensors at native speed. Only the codec itself costs wall time (Table 6: 0.02–0.04 s per 2112-token passage at the balanced point). A production throughput comparison requires a paged-cache backend that embeds the D_4 nested lattice variant of the codec directly into the attention kernel; that engineering is the scope of the upstream PR proposed above and is out of scope for this paper.

9 Related work

9.1 Kakeya problem and harmonic analysis

The Kakeya problem was posed by Kakeya in 1917 [1] as a question about the minimum measure of a region in which a unit segment can be rotated continuously. Besicovitch gave the first nontrivial Kakeya-type constructions in 1919 [2] and 1928 [3], showing sets of arbitrarily small planar measure containing a line segment in every direction. Fefferman's resolution of the ball multiplier problem [4] was the first application connecting Kakeya-type sets to oscillatory-integral bounds; Córdoba [5] developed this further. Bourgain opened the arithmetic-combinatorial era with a nontrivial high-D bound [6]; Wolff introduced the geometric hairbrush argument and pushed the $\mathbb{R}^3$ bound forward [7]; Katz, Łaba, and Tao sharpened the bound for Besicovitch sets in $\mathbb{R}^3$ [8]; Katz and Tao investigated the discretized setting [9]. Tao surveyed the landscape through approximately 2007 [10]. Dvir resolved the finite-field analogue with a one-page polynomial-method proof [11]; Dvir, Kopparty, Saraf, and Sudan extended the technique to list-decoding and extractors [12]. Bennett, Carbery, and Tao posed the multilinear Kakeya conjecture [13], Guth resolved its endpoint case [14], and Guth–Katz [15] resolved the distinct-distances problem in the plane using closely related polynomial-partitioning tools. Hickman, Rogers, and Zhang pushed high-D bounds to near Kakeya-conjectural levels [18]. Zahl investigated restricted Kakeya-type problems [19, 20]. In 2025 Wang and Zahl [21] resolved the three-dimensional Euclidean Kakeya conjecture.

The paper's positioning relies on three specific links in this literature: Guth's multilinear Kakeya endpoint [14], the Brascamp–Lieb equivalence of Bennett–Carbery–Christ–Tao [16], and the direct Brascamp–Lieb-based matrix-concentration proof by Carbery and Valdimarsson [17]. Tropp's user-friendly matrix tail bounds [22] are the concrete form of matrix concentration we rely on.

9.2 Lattice source coding

Zamir and Feder's analysis of lattice quantisation [24] established the basic rate–distortion framework used in this paper. The D_4 root lattice's second-moment optimality in $\mathbb{R}^4$ is a consequence of Conway–Sloane's closest-point analysis [25] and their general treatment [26]. Eyuboglu–Forney [27] formalised the memoryless-source high-rate theory of lattice-bounded codebooks; Forney's trellis-coded modulation work [28] and Erez–Zamir's capacity-achieving lattice construction [29] are closely related in spirit. Chee, Cai, Kuleshov, and De Sa [30] applied lattice codebooks to weight quantisation for LLMs via QuIP#; our codec applies the same structural idea at the KV-cache level with the specific combination of unit-norm factorisation, Hadamard rotation, per-vector adaptive scale, and D_4 closest-point.

9.3 Randomised numerical linear algebra

The Halko–Martinsson–Tropp randomised SVD [23] closes the Brascamp–Lieb to matrix-concentration arc in numerical form. It does not appear in the D_4 nested lattice variant (the RSVD-based v1.3 codec used it extensively [33]), but it is cited because the proof of Proposition 2.2 rests

on matrix-concentration bounds that are derivatives of Tropp's matrix Chernoff inequality [22], whose sharpest form goes through Carbery–Valdimarsson [17].

9.4 KV-cache compression

Token eviction. H_2O [42] and Scissorhands [43] propose attention-score-driven token eviction at test time; StreamingLLM [44] preserves a small prefix of attention-sink tokens exactly and streams the rest. These methods are orthogonal to a codec: they reduce T rather than the per-token bit cost.

Per-channel quantisation. KIVI [40] is a per-channel asymmetric 2-bit quantiser on K combined with per-token on V. Its rotation-free structure sits outside the Hadamard-rotation stack of TurboQuant and KakeyaLattice.

Hadamard-based codecs. *Cache Me If You Must* [41] applies Hadamard rotation combined with adaptive per-token bit-width allocation. TurboQuant [34, 35] provides the production-deployed Hadamard + Lloyd-Max + norm-correction stack we benchmark against (§8); follow-up vLLM PRs include [36, 37, 38] with NIAH long-context validation in [39].

Trained low-rank methods. Dynamic Memory Compression [45] and LESS [46] learn low-rank projections during training; they are orthogonal to the post-hoc codec line studied here.

Prior work in this line. [33] is the prior RSVD + Lloyd-Max codec in our repository. Its rank cap $r = D/2$ relied on a conjectural Kakeya-maximal-function lower bound in dimension $D \geq 4$; the present codec replaces the conjectural bound with the unconditional D_4 shaping bound of Proposition 2.2.

9.5 LLM inference infrastructure

vLLM [48] is the reference serving stack used throughout this paper; its prefix-caching and block-manager abstractions make snapshot-mode measurement possible. The transformers [49] library supplies the weights and tokenisers. FlashAttention [50] is the default attention kernel in all forward passes reported here.

10 Conclusion

We have presented KakeyaLattice, a Zamir–Feder D_4 nested-lattice codec whose shaping gain over scalar quantisation is bounded unconditionally by $G(\mathbb{Z}^4)/G(D_4) \approx 0.919$ via lattice density. Placing the construction inside the Kakeya–Brascamp–Lieb–Tropp dependency chain (§2) grounds the design in a literature-traceable framework: multilinear Kakeya [14] $\rightarrow$ Brascamp–Lieb [16] $\rightarrow$ matrix concentration [22] $\rightarrow$ lattice second-moment optimality [26], with the three-dimensional Kakeya case [21] at the chain's deep end and our codec at its shallow end. The rate–distortion boundary argument of Proposition 2.2 is unconditional at the codec's 4-dimensional block structure, in contrast to the conditional lower bound of our prior work [33].

The measured results for the D_4 nested lattice variant are consistent across four transformer families. On twelve head-to-head pairs at three matched bit points the codec wins universally on K-MSE (12/12), at nine of twelve pairs on $|\Delta\text{ppl}|$, and at all four models at the aggressive $\sim 3.6\times$ compression point. The theoretical shaping-gain prediction of 0.919 is recovered to within $\sim 1\%$ on every tested model in three independent measurement environments. At 128k context the codec reduces the combined KV cache by $2.5\times$–$2.8\times$ at a balanced operating point and $1.66\times$–$1.73\times$ at a near-lossless operating point.

The E_8 nested lattice variant extends these measured results by a further +1.3 to +2.0 dB per-layer K-MSE gain and a 28–53 % $|\Delta\text{ppl}|$ reduction across the same four model families in live vLLM in-forward no-boundary at $n = 32$ passages with 95% CI. The E_8 empirical gain exceeds the +0.29 dB theoretical minimum 4–6×: the two-coset structure of E_8 handles coarse-quantisation outliers better than D_4's single-flip parity correction, so the advantage grows super-linearly under cross-layer error accumulation. Long-context retrieval (Needle-in-a-Haystack at 16k context) is preserved on Qwen3-4B and Gemma-4-E4B at every E_8 operating point. The cost is a fixed +32 bits per vector (−5 to −7 % CR at iso-Q) and 1.56–1.80 × encode latency.

Our positioning (§8) is as an unconditional ≈ 8% improvement to an existing production-deployed KV kernel, exposed as an additive upstream PR. The change is a single per-block quantiser substitution — D_4 closest-point in place of int·per-coord uniform — inside an otherwise unchanged engineering stack, and the measured gain is architecture-independent across four open-source model families.

Reproducibility. All code, unit tests, per-model JSON measurements, the four per-architecture attention hooks, and the multi-model head-to-head harness are released under Apache-2.0 in the `kakeyalattice` package. The canonical codec classes are `kakeyalattice.V14KakeyaZamirLatticeGPU` (D_4 nested lattice variant) and `kakeyalattice.V15KakeyaZamirE8GPU` (E_8 nested lattice variant); both implement the same `.roundtrip(x)` interface parameterised on the lattice choice. The full reproducibility manifest is in Appendix A; the 128k storage report is at `reports/v1_4_release/kv_128k` and the E_8 four-model evaluation is at `reports/v1_5_release/V15_FULL_4MODEL_REPORT.md`.

A note on the distortion metric. Throughout this paper we report reconstruction MSE and its relative normalisation rather than inner-product distortion. For any unit-norm query q, $\left|q^\top(k-\hat{k})\right| \le \left\|k-\hat{k}\right\|$, so an MSE bound is a conservative upper bound on the attention-logit perturbation that matters for downstream quality. The measured $|\Delta\text{ppl}|$ and top-1 agreement corroborate that the MSE proxy is tight in practice.

References

[1] Kakeya, S. Some problems on maxima and minima regarding ovals. *Tohoku Science Reports*, 6:71–88, 1917.

[2] Besicovitch, A. S. On Kakeya's problem and a similar one. *Mathematische Zeitschrift*, 27:312–320, 1928 (original work 1919).

[3] Besicovitch, A. S. On the fundamental geometrical properties of linearly measurable plane sets of points. *Mathematische Annalen*, 98:422–464, 1928.

[4] Fefferman, C. The multiplier problem for the ball. *Annals of Mathematics*, 94(2):330–336, 1971.

[5] Córdoba, A. The Kakeya maximal function and the spherical summation multipliers. *American Journal of Mathematics*, 99(1):1–22, 1977.

[6] Bourgain, J. Besicovitch type maximal operators and applications to Fourier analysis. *Geometric and Functional Analysis*, 1(2):147–187, 1991.

[7] Wolff, T. An improved bound for Kakeya type maximal functions. *Revista Matemática Iberoamericana*, 11(3):651–674, 1995.

[8] Katz, N. H., Łaba, I., Tao, T. An improved bound on the Minkowski dimension of Besicovitch sets in $\mathbb{R}^3$. *Annals of Mathematics*, 152(2):383–446, 2000.

[9] Katz, N. H., Tao, T. Some connections between Falconer's distance set conjecture and sets of Furstenburg type. *New York Journal of Mathematics*, 7:149–187, 2001.

[10] Tao, T. From rotating needles to stability of waves: emerging connections between combinatorics, analysis, and PDE. *Notices of the AMS*, 48(3):294–303, 2001.

[11] Dvir, Z. On the size of Kakeya sets in finite fields. *Journal of the American Mathematical Society*, 22(4):1093–1097, 2009.

[12] Dvir, Z., Kopparty, S., Saraf, S., Sudan, M. Extensions to the method of multiplicities, with applications to Kakeya sets and mergers. *SIAM Journal on Computing*, 42(6):2305–2328, 2013.

[13] Bennett, J., Carbery, A., Tao, T. On the multilinear restriction and Kakeya conjectures. *Acta Mathematica*, 196(2):261–302, 2006.

[14] Guth, L. The endpoint case of the Bennett–Carbery–Tao multilinear Kakeya conjecture. *Acta Mathematica*, 205(2):263–286, 2010.

[15] Guth, L., Katz, N. H. On the Erdős distinct distances problem in the plane. *Annals of Mathematics*, 181(1):155–190, 2015.

[16] Bennett, J., Carbery, A., Christ, M., Tao, T. The Brascamp–Lieb inequalities: finiteness, structure and extremals. *Geometric and Functional Analysis*, 17(5):1343–1415, 2008.

[17] Carbery, A., Valdimarsson, S. I. The endpoint multilinear Kakeya theorem via the Borsuk–Ulam theorem. *Journal of Functional Analysis*, 264(7):1643–1663, 2013.

[18] Hickman, J., Rogers, K. M., Zhang, R. Improved bounds for the Kakeya maximal conjecture in higher dimensions. *American Journal of Mathematics*, 144(6):1511–1560, 2022.

[19] Zahl, J. On the Wolff circular maximal function. *Illinois Journal of Mathematics*, 56(4):1281–1295, 2012.

[20] Zahl, J. A new bound on Kakeya sets in $\mathbb{R}^3$. *arXiv preprint arXiv:2104.13528*, 2022.

[21] Wang, H., Zahl, J. Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions. *arXiv preprint arXiv:2502.17655*, 2025.

[22] Tropp, J. A. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012.

[23] Halko, N., Martinsson, P.-G., Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011.

[24] Zamir, R., Feder, M. On lattice quantization noise. *IEEE Transactions on Information Theory*, 42(4):1152–1159, 1996.

[25] Conway, J. H., Sloane, N. J. A. Fast quantizing and decoding algorithms for lattice quantizers and codes. *IEEE Transactions on Information Theory*, 28(2):227–232, 1982.

[26] Conway, J. H., Sloane, N. J. A. *Sphere Packings, Lattices and Groups*, 3rd ed. Springer, 1999.

[27] Eyuboglu, M. V., Forney, G. D. Lattice and trellis quantization with lattice- and trellis-bounded codebooks—high-rate theory for memoryless sources. *IEEE Transactions on Information Theory*, 39(1):46–59, 1993.

[28] Forney, G. D. Coset codes—part I: introduction and geometrical classification. *IEEE Transactions on Information Theory*, 34(5):1123–1151, 1988.

[29] Erez, U., Zamir, R. Achieving $1/2 \log(1+\mathrm{SNR})$ on the AWGN channel with lattice encoding and decoding. *IEEE Transactions on Information Theory*, 50(10):2293–2314, 2004.

[30] Chee, J., Cai, Y., Kuleshov, V., De Sa, C. QuIP#: Even better LLM quantization with Hadamard incoherence and lattice codebooks. *ICML*, 2024.

[31] Tseng, A., Chee, J., Sun, Q., Kuleshov, V., De Sa, C. QuIP: 2-bit quantization of large language models with guarantees. *NeurIPS*, 2023.

[32] Li, A. KakeyaLattice v1.4: the canonical implementation. Open-source release, *LLM-KV–Cache-compress*, tag `v1.4`, April 2026. Python class `kakeyaturbo_py.V14KakeyaZamirLatticeGPU`; multi-model measurement harness `benchmarks/multimodel_v14_vs_tq.py`; four per-architecture snapshot hooks in `vllm_backend/kakeya_v1_3_ppl/snapshot_hook.py`.

[33] Li, A. Randomized Kakeya Skeletons for LLM KV Cache Compression: Algorithm and Rate–Distortion Boundary. *LLM-KV–Cache-compress v1.3 paper*, April 2026. (Superseded by this paper at tag `v1.4`.)

[34] vibhavagarwal5 *et al.* TurboQuant: 2-bit KV cache compression with 4x capacity. *vLLM merged attention backend*, 2026. (Merged into vllm-project/vllm main on 2026-04-15.)

[35] vibhavagarwal5. [Attention Backend] TurboQuant: 2-bit KV cache compression with 4x capacity. *vllm-project/vllm Pull Request #38479*, 2026. https://github.com/vllm-project/vllm/pull/38479.

[36] Alistarh, D. [Attention] TurboQuant: remove redundant random signs, add prior art attribution. *vllm-project/vllm Pull Request #40194*, 2026.

[37] aditi-amd. [ROCm] Fix TurboQuant on ROCm: backend routing, flash-attn compat. *vllm-project/vllm Pull Request #39953*, 2026.

[38] mgoin. Fix TURBOQUANT backend selection in `cuda.py`. *vllm-project/vllm Pull Request #40060*, 2026.

[39] huangzhilin-hzl. [TurboQuant] Add NIAH long-context eval for TurboQuant KV cache. *vllm-project/vllm Pull Request #40122*, 2026.

[40] Liu, Z., Yuan, J., Jin, H., Zhong, S., Xu, Z., Braverman, V., Chen, B., Hu, X. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. *ICML*, 2024.

[41] Shutova, E. *et al.* Cache Me If You Must: Adaptive KV Cache Quantization for LLMs. *arXiv preprint arXiv:2501.19392*, 2025.

[42] Zhang, Z., Sheng, Y., Zhou, T., Chen, T., Zheng, L., Cai, R., Song, Z., Tian, Y., Ré, C., Barrett, C., Wang, Z., Chen, B. H_2O: Heavy-hitter oracle for efficient generative inference of large language models. *NeurIPS*, 2023.

[43] Liu, Z., Desai, A., Liao, F., Wang, W., Xie, V., Xu, Z., Kyrillidis, A., Shrivastava, A. Scissorhands: Exploiting the persistence of importance hypothesis for LLM KV cache compression at test time. *NeurIPS*, 2023.

[44] Xiao, G., Tian, Y., Chen, B., Han, S., Lewis, M. Efficient streaming language models with attention sinks. *ICLR*, 2024.

[45] Nawrot, P., Lancucki, A., Chochowski, M., Tarjan, D., Ponti, E. M. Dynamic memory compression: Retrofitting LLMs for accelerated inference. *arXiv preprint arXiv:2403.09636*, 2024.

[46] Dong, H., Yang, X., Zhang, Z., Wang, Z., Chi, Y., Chen, B. Get more with LESS: Synthesizing recurrence with KV cache compression for efficient LLM inference. *ICML*, 2024.

[47] DeepSeek-AI. DeepSeek-V3 Technical Report. *arXiv preprint arXiv:2412.19437*, 2024.

[48] Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., Stoica, I. Efficient memory management for large language model serving with PagedAttention. *SOSP*, 2023.

[49] Wolf, T. *et al.* Transformers: State-of-the-art natural language processing. *EMNLP (Demo)*, 2020.

[50] Dao, T., Fu, D. Y., Ermon, S., Rudra, A., Ré, C. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. *NeurIPS*, 2022.

A Reproducibility manifest

The full multi-model benchmark for *both codecs* is reproducible via the rigorous evaluation harness introduced with the v1.5 release:

```
cd LLM-KV--Cache-compress
pip install -e kakeyalattice
pip install -e vllm_backend
export VLLM_ENABLE_V1_MULTIPROCESSING=0 KAKEYA_SNAPSHOT_QWEN3=1

# PPL / MSE / CR (v1.4 + v1.5 + TurboQuant at matched Q / b):
python benchmarks/rigorous_eval.py \
    --model-path <HF-id> --model-name <short>_nobdry \
    --mode inforward --no-boundary \
    --q-values 4,10 --v15-q-values 4,10 --tq-b-values 3 \
    --kv-modes KV \
    --ctx-len 2048 --n-eval 64 --n-passages 32 \
    --out-dir reports/v1_5_release

# TurboQuant b=2 guardrail (requires boundary=2 to boot):
python benchmarks/rigorous_eval.py \
    --model-path <HF-id> --model-name <short>_tqb2_bdry2 \
    --mode inforward --boundary-size 2 \
    --q-values "" --v15-q-values "" --tq-b-values 2 \
    --kv-modes KV \
    --ctx-len 2048 --n-eval 64 --n-passages 32 \
    --out-dir reports/v1_5_release

# Pure codec latency (no model needed):
python benchmarks/e8_latency_benchmark.py --n-iters 500

# NIAH long-context retrieval:
python benchmarks/niah_eval.py \
```

```
    --model-path <HF-id> --model-name <short> \
    --mode inforward --boundary-size 2 --n-trials 3 \
    --ctx-lengths 4096,8192,16384 --depths 0.1,0.5,0.9 \
    --q-values 4,10 --v15-q-values 4,10 --tq-b-values 2,3 \
    --out-dir reports/v1_5_release/niah

# Frozen sha256 parity (bit-level regression gate):
python benchmarks/e8_parity_and_smoke.py
```

Replace `<HF-id>` with one of `Qwen/Qwen3-4B`, `deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B`, `google/gemma-4-E4B`, or `zai-org/GLM-4-9B-Chat` (add `--trust-remote-code` for the last). Raw per-passage JSON and full stdout logs are committed under `reports/v1_5_release/` (v1.5 data) and `reports/v1_4_release/` (v1.4 frozen data).

B Canonical naming

- **Project name**: KakeyaLattice
- **Python package**: `kakeyalattice`
- **v1.4 codec (D_4)**:
  - spoken / written: "v1.4 kakeya zamir lattice GPU"
  - with parameter: e.g. "v1.4 $Q = 152$"
  - class: `V14KakeyaZamirLatticeGPU`
  - module: `kakeyalattice.v1_4_kakeya_zamir_lattice_gpu`
  - release tag: `v1.4` (commit `6b02711`)
- **v1.5 codec (E_8)**:
  - spoken / written: "v1.5 kakeya zamir E8 GPU"
  - with parameter: e.g. "v1.5 $Q = 10$"
  - class: `V15KakeyaZamirE8GPU`
  - module: `kakeyalattice.v1_5_kakeya_zamir_e8_gpu`
  - release tag: `v1.5` (commit at the time of paper release)

C Canonical operating points

For $D = 128$ (head_dim of Qwen3-4B, DeepSeek-R1-Distill-1.5B, and GLM-4-9B-Chat):

	D_4 nested lattice variant		E_8 nested lattice variant		
q_{range}	bits	CR	bits	CR	Role
152	1088	1.88×	1104	1.86×	Near-lossless
76	960	2.13×	976	2.10×	
38	832	2.46×	848	2.41×	D_4 **balanced**
19	704	2.91×	720	2.84×	
10	576	3.56×	608	3.37×	E_8 **balanced**
5	448	4.57×	464	4.41×	
4	416	4.92×	448	4.57×	Aggressive compression
2	320	6.40×	336	6.10×	Extreme compression

For Gemma-4-E4B's heterogeneous head_dim (20 sliding-attention layers with $d_h = 256$, 4 full-attention layers with $d_h = 512$), both codecs emit per-layer bit budgets via the `LatticeCodebook` dispatcher in `kakeyalattice.lattice_codebooks`; the aggregated 128k storage figures in Table 5 use those per-layer values.

Deployment recommendation. Use E_8 **Q=**10 (CR 3.37×, $|\Delta\text{ppl}|$ typically 1.6–6.9%) as the default head-of-line operating point. Use E_8 Q=4 (CR 4.57×) when aggressive compression is non-negotiable and $|\Delta\text{ppl}|$ of 6–32% is acceptable on the target model family. Use E_8 Q=152 (CR 1.86×, $|\Delta\text{ppl}| < 1\%$) when near-lossless is required. Stay on the D_4 nested lattice variant for latency-critical paths where the 1.56–1.80× E_8 encode cost is unacceptable.